

GUSTAVO ARAUJO

AI Inference · C++ Systems · Technical Leadership

Lead Software Engineer with 9+ years of experience. Currently shipping production AI models on-device, without server and with GPU acceleration, and leading a 6-person team to deliver transcription and voice generation models. Before that I also worked for 5 years building and re-architecting the C/C++ PCIe verification software at Cadence, where I led the Verification IP and earned three patents in machine learning for hardware verification. Electrical Engineering degree with a Computer Science minor, plus a graduate degree in AI/ML.

SELECTED HIGHLIGHTS

- Patent: [Reinforcement learning for test generation](#) (US 12,099,791 · Granted)
- Patent: [Sequence-number monitor synchronization](#) (US 12,348,403 · Granted)
- Patent: Robust data-stream identification (Filed)
- Speaker: [PCI-SIG U.S. Developers Conference](#) (search Gustavo Araujo)
- Blog: [Insights Into the Evolutions and Optimizations of PCIe 6.0](#)

PROFESSIONAL EXPERIENCES

[Tether Operations Limited](#) - Remote (Feb'25-current)

Worked on [QVAC](#), an open-source, local AI Inference ecosystem (C++, JS)

Roles: Lead Software Engineer, C++ Developer

- Led 6-person team owning on-device speech stack (ASR, TTS)
- Converted several models to GGML backend (Parakeet, Supertonic, Chatterbox)
- Provided streaming endpoints for transcription (including Whisper) and TTS
- Achieved faster than real time inference in all windows, linux, mac, iOS, android
- Halved model sizes with negligible quality loss through GGUF quantization

[Cadence Design Systems](#) - Belo Horizonte, Brazil (Apr'20-Feb'25)

Software development (C/C++) for hardware verification (SV) of PCIe protocol

Roles: Lead Software Engineer, Software Engineer II and I, and Intern

- Created databases and trained AI LLMs to create a chat bot for VIP
- Rearchitected the software for implementation of PCIe 6.0 and PCIe 7.0
- Planned PCIe new features, lead R&D, organized feature customer support

[Consulting and Electrical Projects](#) - Belo Horizonte, Brazil (Jun'17-Dec'19)

University's engineering and software development company

Roles: Project Director, Project Manager, Developer

- Lead a team of 5 managers and 15 developers developing websites
- Introduced SCRUM, improving NPS by 70% and reducing overdue projects by 42%

ACADEMIC EDUCATION

[Federal University of Minas Gerais \(UFMG\)](#) - Belo Horizonte, Brazil. GPA: 4.85/5.0

Bachelor's in Electrical Engineering with minor in Comp. Science (2017 – 2021)

Final project: [Smartphone app for portable microscope capable of identifying Mycobacterium tuberculosis bacilli](#). Android development in Java ([repository](#))

[PUC Minas](#) - Belo Horizonte, Brazil

Graduation in Artificial Intelligence/Machine Learning (Jul'22 – Jun'23)

[Wayne State University](#) - Detroit, USA. GPA: 4.0/4.0

Exchange Program in Computer Science (Jan'20 – Apr'20)

CONTACT INFO

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Check [my website](#)

SKILLS

[Programming Languages](#)

C · C++ · Python · JS

[AI & Inference](#)

On-device inference · Model quantization · Text to speech (TTS) · Neural Machine Translation (NMT) · Automatic speech recognition (ASR) · GGML · ONNX · Dataset curation · Huggingface

[Leadership & Delivery](#)

Team leadership · Agile / SCRUM · Roadmap & Prioritization · Mentoring · Training

LANGUAGES

Portuguese (native)

English (fluent)

Spanish (intermediate)

AWARDS

Second highest overall grade among graduating students at UFMG (Oct'21)

BEYOND WORK

- Professional Certificate for Game Designer and Unity Developer at [EBAC](#)
- Writer of ["O Tempo Fala"](#), a time management book